EASY · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Inference from sample statistics and margin of error

01

9 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

C

C. Plausible values of the percent of the students at the school who support a menu change are between 32.5% and 43.5%.

Choice C is correct. It’s given that an estimated 38% of sampled students at the school were in support of a menu change, with a margin of error of 5.5%. It follows that the percent of the students at the school who support a menu change is 38% plus or minus 5.5%. The lower bound of this estimation is $38 - 5.5$, or 32.5%. The upper bound of this estimation is $38 + 5.5$, or 43.5%. Therefore, plausible values of the percent of the students at the school who support a menu change are between 32.5% and 43.5%.

Choice A is incorrect. This is the percent of the sampled students at the school who support a menu change. Choices B and D are incorrect and may result from misinterpreting the margin of error.

Q2

A

A. The mean weight of all handbags made by the company on that day is between 27.78 oz and 27.82 oz.

Choice A is correct. It's given that the estimated mean weight of all handbags made by the company on a particular day is 27.8 oz, with an associated margin of error of 0.02 oz. It follows that plausible values for the mean weight are between $(27.8 - 0.02)$ oz and $(27.8 + 0.02)$ oz. Therefore, the most plausible conclusion is that the mean weight of all handbags made by the company on that day is between 27.78 oz and 27.82 oz.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

C

C. It is between 34 and 40.

Choice C is correct. It's given that the mean of the data from a random sample of a population is 37, with an associated margin of error of 3. The most appropriate conclusion that can be made is that the mean of the entire population will fall between 37, plus or minus 3. Therefore, the population mean is between $37-3=34$ and $37+3=40$.

Choice A is incorrect. While it's an appropriate conclusion that the population mean is as low as $37-3$, or 34, it isn't appropriate to conclude that the population mean is less than 34. Choice B is incorrect. While it's an appropriate conclusion that the population mean is as high as $37+3$, or 40, it isn't appropriate to conclude that the population mean is greater than 40. Choice D is incorrect. It isn't an appropriate conclusion that the population mean is less than 34 or greater than 40.

Q4

A

A. It is between $\$4.15$ and $\$4.31$.

Choice A is correct. It's given that the mean price of a carton of grape tomatoes in Utah was estimated to be $\$4.23$, with an associated margin of error of $\$0.08$. It follows that plausible values for this mean price are between $\$4.23$ - $\$0.08$ and $\$4.23+\$0.08$. Therefore, it's plausible that the mean price of a carton of grape tomatoes for all locations that sell this product in Utah is between $\$4.15$ and $\$4.31$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

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Answer not provided in source data — see rationale below.

Choice C is correct. Because a random sample of the city council was polled, the proportion of the sample who supported the bill is expected to be approximately equal to the proportion of the total city council who supports the bill. Since 6 of the 20 polled, or 30%, supported the bill, it can be estimated that 50×0.3, or 15, city council members support the bill.

Choice A is incorrect. This is the number of city council members in the sample who supported the bill. Choice B is incorrect and may result from a computational error. Choice D is incorrect. This is the number of city council members in the sample of city council members who were not polled.

Q6

B

B. 320

Choice B is correct. It's given that from the sample of 20 employees at the company, 16 of the employees are enrolled in exactly three professional development courses this year. Since $\frac{16}{20}$ is equal to 0.80, or $\frac{80}{100}$, it follows that 80% of the employees in the sample are enrolled in exactly three professional development courses this year. Therefore, the best estimate for the percentage of employees at the company who are enrolled in exactly three professional development courses this year is 80%. It's given that there are a total of 400 employees at the company. Therefore, the best estimate of the number of employees at the company who are enrolled in exactly three professional development courses this year is $\frac{80}{100}400$, or 320.

Choice A is incorrect. This is the number of employees from the sample who aren't enrolled in exactly three professional development courses this year.

Choice C is incorrect. This is the number of employees who weren't selected for the sample.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7 A

A. 11

Choice A is correct. It's given that 20% of the students surveyed responded that they intend to enroll in the study program. Therefore, the proportion of students in Spanish club who intend to enroll in the study program, based on the survey, is 0.20. Since there are 55 total students in Spanish club, the best estimate for the total number of these students who intend to enroll in the study program is 550.20, or 11.

Choice B is incorrect. This is the best estimate for the percentage, rather than the total number, of students in Spanish club who intend to enroll in the study program.

Choice C is incorrect. This is the best estimate for the total number of Spanish club students who do not intend to enroll in the study program.

Choice D is incorrect. This is the total number of students in Spanish club.

Q8 B

B. 2,100

Choice B is correct. Let x be the number of people in the entire town that would be expected to name chocolate. Since the sample of 50 people was selected at random, it is reasonable to expect that the proportion of people who named chocolate as their favorite ice-cream flavor would be the same for both the sample and the town population. Symbolically, this can be expressed as $\frac{7}{50} = \frac{x}{14{,}878}$. Using cross multiplication, $7 \times 14{,}878 = x \times 50$; solving for x yields 2,083. The choice closest to the value of 2,083 is choice B, 2,100.

Choices A, C, and D are incorrect and may be the result of errors when setting up the proportion, solving for the unknown, or incorrectly comparing the choices to the number of people expected to name chocolate, 2,083.

Q9 C

C. 13,500 to 14,500

Choice C is correct. The mean of the 5 samples is $\frac{51+59+45+52+73}{5}=56$ trees per acre.

The best estimate for the total number of trees in the forest is the product of the mean number of trees per acre in the sample and the total number of acres in the forest. This is $(56)(253) = 14,168$, which is between 13,500 and 14,500.

Choice A is incorrect and may result from multiplying the minimum number of trees per acre in the sample, 45, by the number of acres, 253. Choice B is incorrect and may result from multiplying the median number of trees per acre in the sample, 52, by the number of acres, 253. Choice D is incorrect and may result from multiplying the maximum number of trees per acre in the sample, 73, by the number of acres, 253.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q5 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

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